

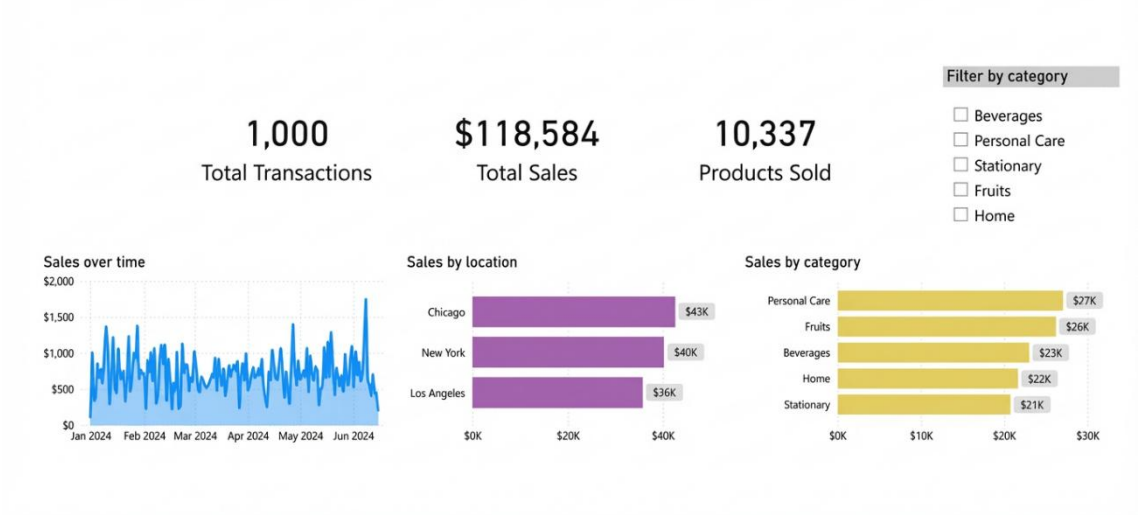
Retail Sales Performance Analysis

Business Intelligence & Data Analysis Project

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Tools: Power BI | SQL | Excel

Supermarket Performance



Executive Summary: “Comprehensive analysis of 1,000 transactions to identify revenue optimization opportunities in three major locations (Chicago, NY, LA).”

Key Findings (Insights):

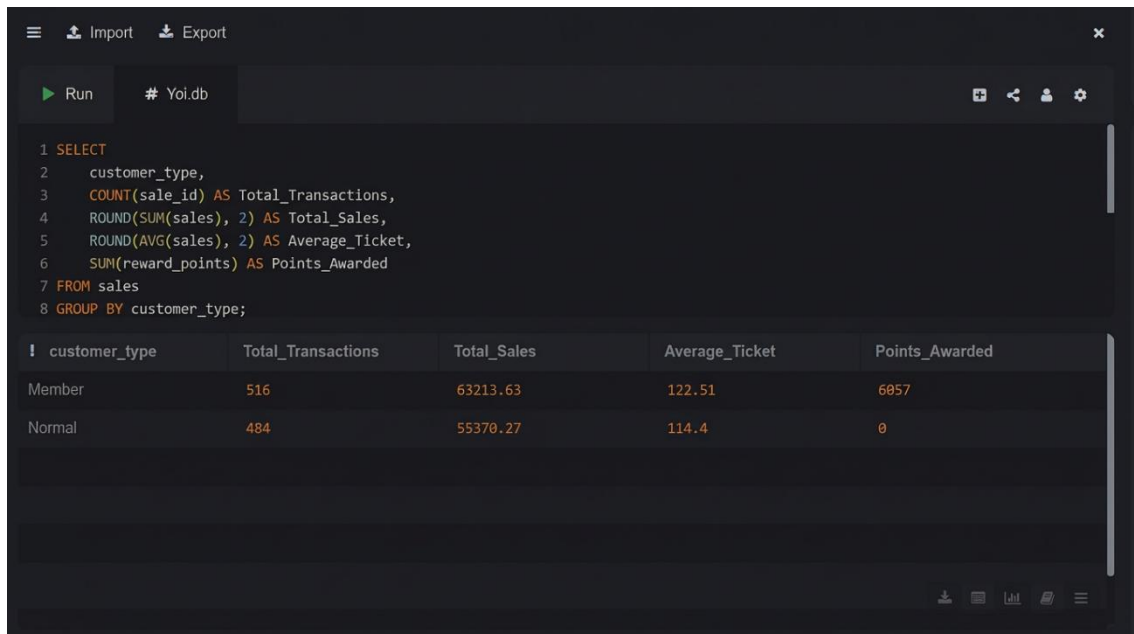
- Chicago Leadership:** The Chicago office leads in revenue with \$42,584.71, surpassing New York (\$40,226.93) by a margin of 5%.
- High Season:** A peak in sales volume was detected in June.
- Top Categories:** “Personal Care” and “Fruits” account for 45% of total revenue, representing the pillars of the business.

Recommendations:

Investigate poor performance in Los Angeles and launch localized promotional campaigns.

Data Extraction and Analysis with SQL

I used SQL to query the raw database, calculate efficiency metrics, and prepare the data for visualization.



The screenshot shows a SQL IDE interface with a dark theme. At the top, there are menu items: 'Run', 'Import', and 'Export'. Below the menu, there's a tab labeled '# Yoi.db'. The main area contains a SQL query:

```
1 SELECT
2   customer_type,
3   COUNT(sale_id) AS Total_Transactions,
4   ROUND(SUM(sales), 2) AS Total_Sales,
5   ROUND(AVG(sales), 2) AS Average_Ticket,
6   SUM(reward_points) AS Points_Awarded
7 FROM sales
8 GROUP BY customer_type;
```

Below the query, the results are displayed in a table with the following columns: customer_type, Total_Transactions, Total_Sales, Average_Ticket, and Points_Awarded. The data is grouped by customer_type, showing results for 'Member' and 'Normal'.

customer_type	Total_Transactions	Total_Sales	Average_Ticket	Points_Awarded
Member	516	63213.63	122.51	6057
Normal	484	55370.27	114.4	0

Technical Explanation:

- **Aggregation:** Using GROUP BY and SUM to calculate total sales by category.
- **Calculation of Metrics:** Creation of calculated fields such as Average Ticket and Tax Ratio.
- **Sorting:** Using ORDER BY to quickly identify the products with the greatest impact.

Data Cleaning, Transforming and Loading (ETL)

Data normalization process using Power Query to ensure the integrity of the analysis.

sale_id	branch	city	customer_type	gender	product_name	product_category
1	A	New York	Member	Male	Conditioner	Personal Care
2	B	Los Angeles	Normal	Female	Notebook	Stationery
3	A	New York	Member	Female	Apple	Fruits
4	A	Chicago	Normal	Male	Cleaner	Household
5	B	Los Angeles	Member	Female	Soda	Beverages
6	A	Chicago	Normal	Male	Pen	Stationery
7	A	Chicago	Normal	Male	Shampoo	Personal Care
8	B	Los Angeles	Normal	Female	Detergent	Household
9	A	Chicago	Member	Female	Banana	Fruits
10	B	Los Angeles	Member	Male	Apple	Fruits
11	A	Chicago	Normal	Male	Orange Juice	Beverages
12	B	Los Angeles	Member	Male	Cleaner	Household
13	A	New York	Normal	Female	Detergent	Household
14	A	Los Angeles	Member	Female	Banana	Fruits
15	B	Los Angeles	Normal	Female	Soda	Beverages
16	B	Los Angeles	Normal	Male	Pen	Stationery
17	B	Los Angeles	Member	Female	Conditioner	Personal Care
18	B	Los Angeles	Normal	Male	Banana	Fruits
19	A	New York	Member	Male	Orange Juice	Beverages
20	A	Chicago	Normal	Male	Shampoo	Personal Care
21	B	Los Angeles	Normal	Female	Shampoo	Personal Care
22	B	Los Angeles	Normal	Male	Soda	Beverages
23	A	New York	Member	Female	Soda	Beverages
24	A	Chicago	Normal	Male	Banana	Fruits
25	B	Los Angeles	Member	Male	Orange Juice	Beverages
26	B	Los Angeles	Normal	Male	Apple	Fruits
27	A	New York	Member	Male	Detergent	Household
28	A					

Technical Explanation:

- **Temporary Normalization:** Generation of a sequential timeline to enable trend analysis (Time Intelligence).
- **Type Cleaning:** Conversion and validation of data types (Currency, Date, Text) to avoid calculation errors.
- **Filtering:** Removal of noise in raw data before loading into the model.